Modular Digital Twin Architecture for Smart Building Energy Optimization: A Open-Source Approach

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Abstract

This paper presents the design, implementation, and deployment of a modular digital twin (DT) architecture for smart building environments, with a focus on energy optimization and real-time control. Motivated by the growing need for sustainable and efficient urban infrastructure, the proposed solution integrates low-cost IoT sensors, open-source software, and data analytics as an attempt to deliver a scalable and interoperable platform. Experiments were conducted in laboratory settings on a university campus, highlighting the architecture's cost-effectiveness and ease of maintenance. Future work will address comprehensive scalability, enhanced AI integration, and open-source alternatives for visualization.

CCS Concepts

• **Computer systems organization** → *Cloud computing*; **Real-time system architecture**; **Embedded and cyber-physical systems**; • **Software and its engineering** → **Layered systems**; • **Computing methodologies** → *Model verification and validation.*

Keywords

Digital Twin, Smart Cities, Smart Buildings, IoT, Energy Efficiency

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1 Introduction

Smart cities are an emerging topic, frequently associated with the idea of technologies and innovation to increase urban efficiency and sustainability [6]. As an integral component of smart urban development, smart buildings represent a scalable and modular approach that enhances flexibility and efficiency while contributing to the reduction complexity inherent in smart cities, which can be modeled as systems of systems.

By integrating technologies that provide automation, data analysis, and artificial intelligence (AI), it should be easier to establish a basis for a more efficient management of smart city infrastructure. Digital twins (DTs) presents as a unified digital model that collects diverse data from physical systems to create an accurate and comprehensive virtual representation. Although they offer a comprehensive solution, they also entail significant risks, making the achievement of trustworthiness paramount [2]. To serve as a trustworthy and unalterable record of a physical asset's state, integrity is essential, requiring a robust and critical infrastructure.

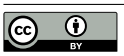
Building on the notion that effective automation initiatives are best implemented at a smaller scale, this paper presents a layered DT approach to address the challenges identified in current IoT deployments. In the Instituto de Ciências Matemáticas e de Computação (ICMC) campus, more than 400 air conditioners are operated manually, accounting for up to 50% of the electricity bill.

2 System Architecture

The architecture developed in this study is built on the initial prototype structure proposed in [8], integrating analytics to optimize energy use. By following a four-layer approach, a common design in digital twin development [5], we bridge the physical and digital worlds with an interface that supports data integration and analytics.

As a real-life component may be defined by different characteristics, the inherent heterogeneity of physical assets presents a challenge to unify their information streams. To overcome this, IoT middlewares are essential, functioning as an abstraction that simplifies the integration of these devices with broader system components, making it easier to manage and process data from disparate sources.

By supporting multiple communication protocols, and interoperable mechanisms that allow connections to existing backend, predictions can be achieved through stored sensor data that serves as the foundation for the predictive modeling phase, exposed through an interactive interface for the user. The overall structure can be seen in Figure 1.



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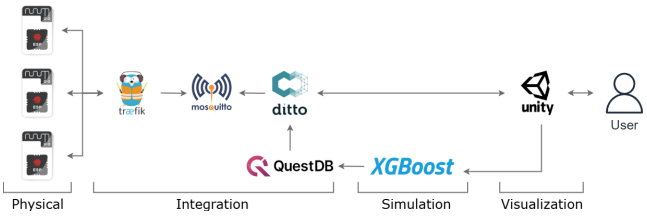


Figure 1: Overall architecture of the digital twin and its technologies.

3 Implementation and Case Study

The architecture is implemented and validated at the ICMC campus [7]. The research group operates two laboratories, one of which constitutes the object of analysis in this case study.

This laboratory, shown in Figure 2, comprises an infrastructure of four servers, each operating continuously. To maintain adequate cooling, air-conditioning units are dedicated to server refrigeration and students comfort, functioning around the clock.

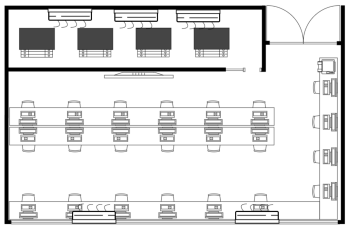


Figure 2: Blueprint of the ICMC lab used as study object.

By deploying sensors throughout the laboratory, monitoring and control of the indoor environment can be achieved. Among the sensors utilized, the DHT22 [1] sensor is selected for its cost-effectiveness and capability to measure both temperature and relative humidity, while the SCT-013 sensor is chosen for its reliable measurement of electrical current.

Integrating ESP-WROOM-32 units enables real-time data transmission through the MQTT protocol [4], thereby supporting data ingestion into the Eclipse Ditto-based twin. Figure 3 depicts the selected sensors and microcontroller. Each sensor node publishes its measurements, such as temperature, humidity, and energy consumption, to designated MQTT topics. Corresponding command topics enable remote configuration and control.

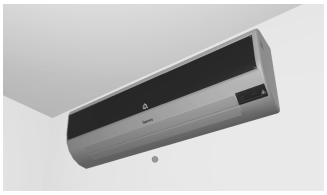


Figure 3: Devices used in the laboratory environment.

A regression model was used to forecast energy consumption (kWh) using time-series methods. The model utilizes lagged values of prior energy usage, temperature, and humidity as predictive features. Validation was conducted using Time-Series Cross-Validation (TimeSeriesSplit), which preserves the temporal order of the data, enabling training on historical observations and evaluation on future data points. Early stopping was employed as a regularization technique to mitigate overfitting. The chosen algorithm - Extreme Gradient Boosting (XGBoost) [3] - builds an ensemble of decision trees that iteratively correct residual errors, presenting robust performance in energy consumption prediction tasks [10].



(a) Physical A/C unit.



(b) Virtual A/C counterpart.

Figure 4: Physical and Virtual Lab A/Cs.

Because digital twins mirror physical assets in real time, an intuitive user interface can be useful. With Unity [9], a real-time 3D engine, users can control the real air conditioning (A/C) unit through its virtual counterpart, while triggering analytics by operating schedules, temperature, and humidity inputs. Figure 4 shows both the virtual and physical components.

4 Results and Discussion

The predictive performance of the XGBoost model was evaluated using unseen temporal data from the laboratory's air-conditioning system. Figure 5 presents a graphical comparison of measured (blue) and predicted (orange) energy consumption values (kWh) during the most recent validation period. The measured values display noticeable peaks and troughs, reflecting the on-and-off cycles of the air-conditioning compressor controlled by a thermostat. In contrast, the predicted values appear smoother, as the model captures overall trends but does not replicate every rapid fluctuation.

To demonstrate how the predictive component of the digital twin can assist in evaluating energy performance under different conditions, three simulation scenarios were executed through the system's web API, shown in Figure 6.

Executed in October 2 of 2025, scenarios 1 and 2 were configured with identical indoor parameters (22°C and 50% relative humidity) and the same operation time (five hours), but during different periods of the day. Scenario 1 was executed in the morning period, resulting in an estimated energy consumption of 2.14 kWh, whereas Scenario 2 was simulated in the afternoon, consuming 2.73 kWh. This difference reflects the higher external temperature during the afternoon, which increases cooling effort to maintain the same indoor setpoint.

Scenario 3 maintained the system active for both periods, and at a higher indoor temperature (24°C), resulting in a total predicted consumption of 3.78 kWh. This higher value compensates for the lower thermal demand, leading to overall energy usage consistent with expected physical behavior.

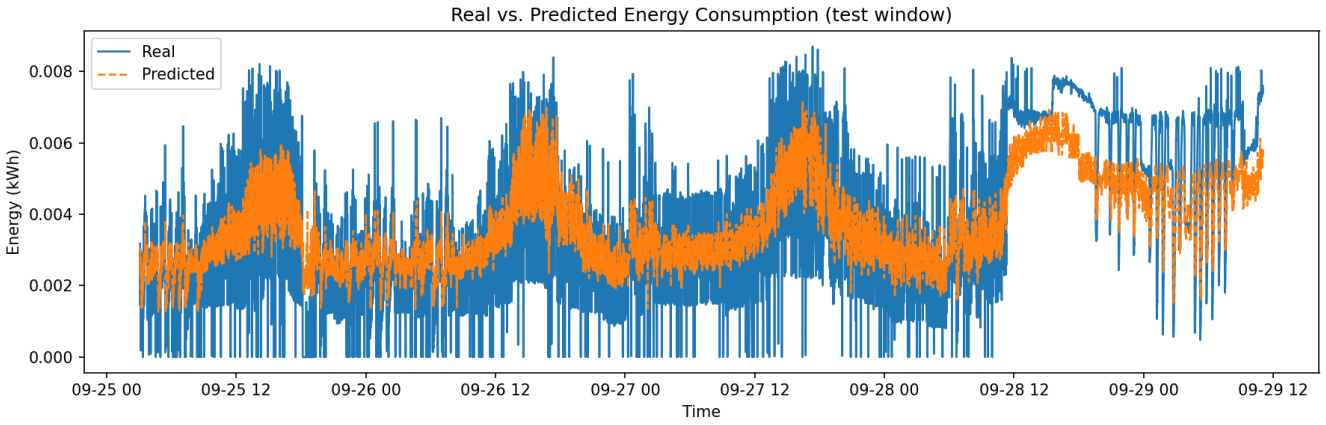


Figure 5: Comparison between measured and predicted energy consumption (kWh) for the validation period. The model captures overall trends and transitions, despite smoothing high-frequency fluctuations.

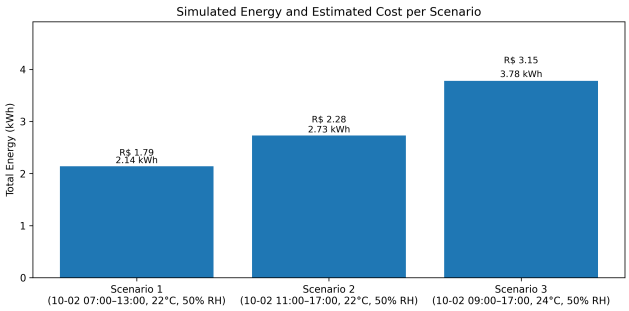


Figure 6: Simulated energy consumption and estimated cost for different user-defined scenarios executed on October 2 - 2025.

These results illustrate how the simulation tool can support energy management strategies in smart buildings. By anticipating consumption and cost under different configurations, managers can identify operational schedules that minimize energy demand while maintaining comfort.

5 Threats to Validity

Experiments were limited to one month. Within this test month, data were collected, but results can vary under different climate conditions. As for the selected regression model, it was trained on a specific dataset. Attaching other variables could potentially remove biases and expand its ability to generalize broadly representative scenarios. Additionally, Unity is not fully open source. While the proprietary licensing model allows its usage under specific revenue thresholds, suitable for this use case, its long-term availability may be subject to changes.

6 Conclusion and Future Work

We presented a modular and integrated digital twin applied to a smart laboratory. The system demonstrated good usability through a 3D representation, interoperability by supporting different sensors, and an overall low-cost approach that leverages open-source software tools followed by non-invasive sensors. By applying it in a laboratory on campus, we fulfill these objectives, strengthening a foundation for smart campus ecosystems, bridging research and practical deployment of digital twins.

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